HAND POSE PREDICTION DRIVEN BY MOTONEURON SYNERGIES

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Transradial Post Stroke Rehabilitation Solutions

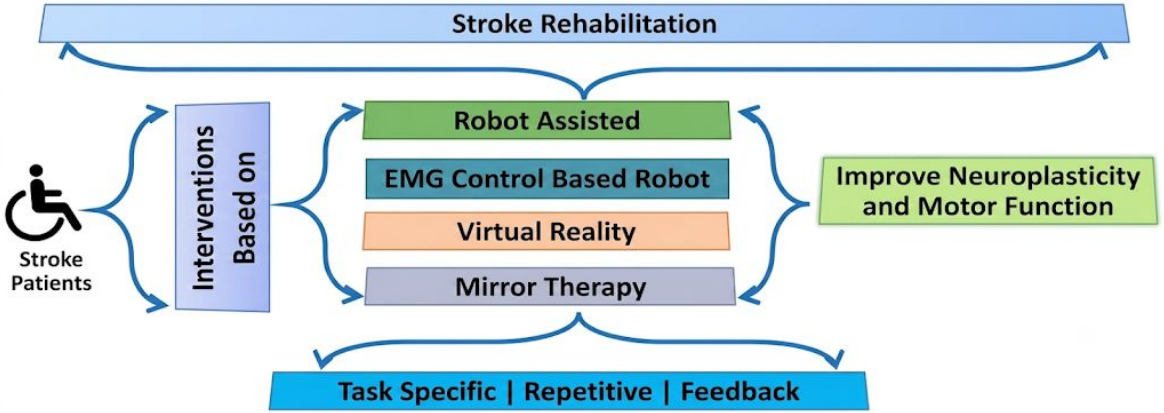
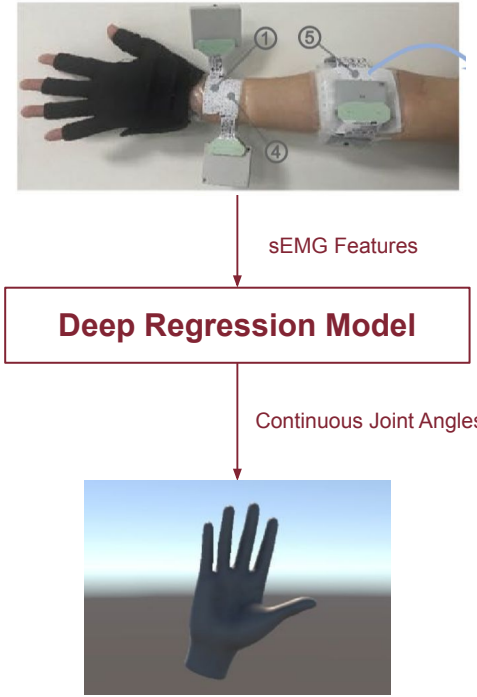


Fig.1 "*Maximizing stroke recovery with advanced technologies: A comprehensive assessment of robot-assisted, emg-controlled robotics, virtual reality, and mirror therapy interventions*" Amin et al., 2024



sEMG ENABLES INTENTION DRIVEN CONTROL

⇒ Real Time Responsiveness < 300 ms

From Traditional Pattern Recognition to HD-sEMG

➔ **Sparse sEMG Pattern Recognition**

- Time/Frequency Features Extraction
- Macroscopic Averaged Representation of Muscle Activity
- Steady Contractions

➔ **High Density sEMG Convolutive Mixing Model**

$$sEMG_j(t) = \sum_{i=1}^{n} (MUST_i(t) * MUAP_{i,j}(t)) + e(t)$$

MUST : *Motor Unit Spike Train*

MUAP : *Motor Unit Action Potential*

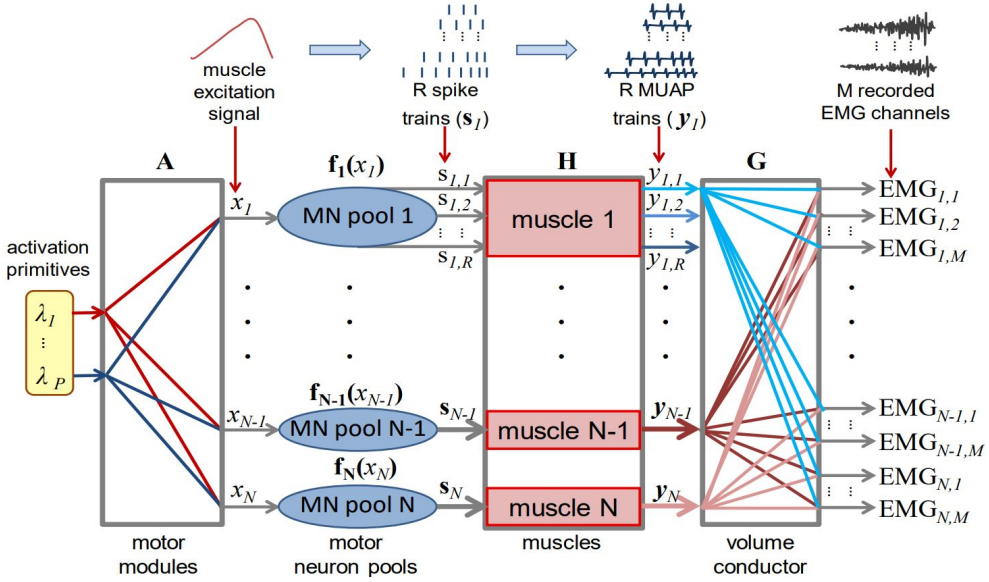


Fig.2 "*Blind source identification from the multichannel surface electromyogram*" Holobar and Farina (2014)

Motor Redundancy and Synergy Hypothesis

The CNS simplifies hand control by grouping Degrees of Freedom into functional synergies.

➔ Muscle Synergies

Extracted via *Non-Negative Matrix Factorization* (NNMF) from sEMG envelopes:

$$V \approx W \cdot H$$

➔ Motoneuron Synergies

Extracted via NNMF from *Motor Unit Smoothed Discharge Rates* (SDRs):

$$S \approx W_{MN} \cdot H_{MN}$$

- Superior Orthogonality
- Higher Expressive Power

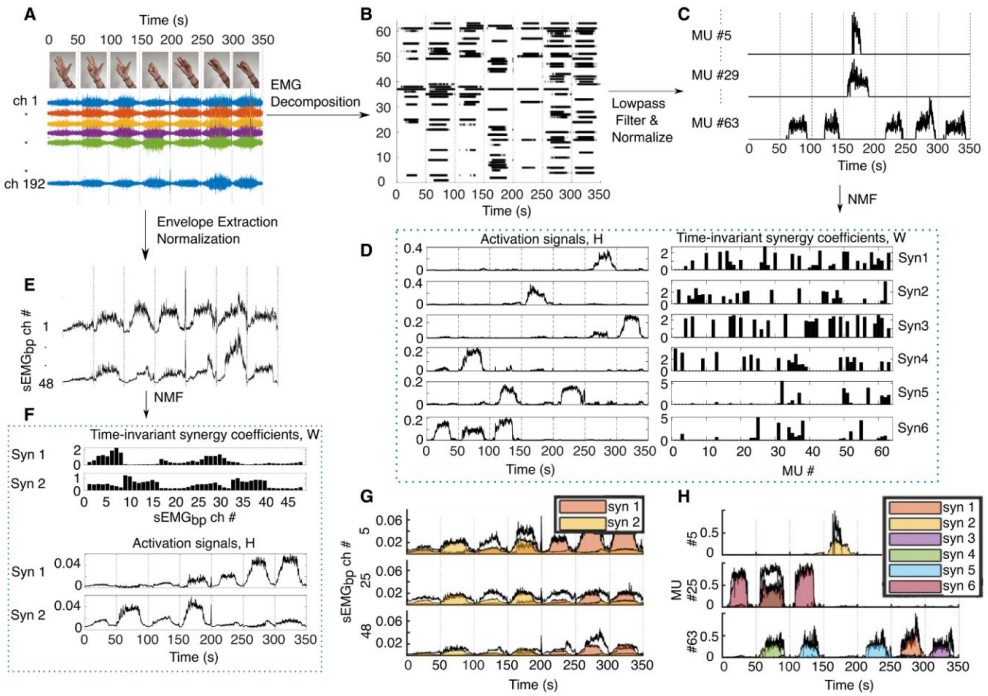


Fig.3 "Merging motoneuron and postural synergies in prosthetic hand design for natural bionic interfacing" Capsi-Morales et. al (2025)

Map Motoneuron Synergies to Hand Pose

Pipeline and Contributions:

- ***Preliminary Experimental Setup***
- ***Open Source Dataset Preprocessing***
- ***Motor Unit Decomposition and MUs Postprocessing***
- ***Motoneuron Synergies***
- ***Synchronization***
- ***Model Architecture/Kinematic Prediction***
- ***Unity Validation/Results***

Dataset: Preliminary Experimental Setup



HD sEMG Acquisition

64 channel HD sEMG Grids

EMG USB2+ Amplifier

OTBioLab+ Software

Kinematics Acquisition

MANUS Quantum Metagloves

Movella Xsens System

Hardware Sync: TTL Pulse via BNC

MANUS Core and Xsens Software



Open Source Dataset

- **Analyzed Dataset**
  - **SEEDS**
  - **HD FW KIN**
    - Clearer Isometric Phases
    - Easier Segmentation
    - Higher Inter-Trial Consistency

	CapgMyo	SEEDS	GrabMyo	Hyser	CEMHSEY	HD-FW KIN
No. Subjects	10–18	25	43	20	13 (GRASP), 6 GESTURE)	10
Amputees?	No	No	No	No	No	No
No. Movements/Gestures	8, 12, 2	13	16	34	7 (GRASP), 11 (GESTURE)	13
No. Electrodes	128 (HD)	128 (HD) +8	28	256 (HD)	320 (HD)	448 (HD)
Kinematics or Force?	No	Kinematics	No	Finger Force	Hand-Gripping & Finger Force (GRASP), No (GESTURE)	Kinematics (5DT Data Glove 14 Ultra)
Multi-Day?	2 days	No	3 days	2 days	11 days	No
Consecutive Days?	No	\	No	Yes	Yes	No

Fig. 4 "A consecutive multi-day high-density surface electromyography dataset comprising 7 grasps and 11 gestures," Yang et al., 2025

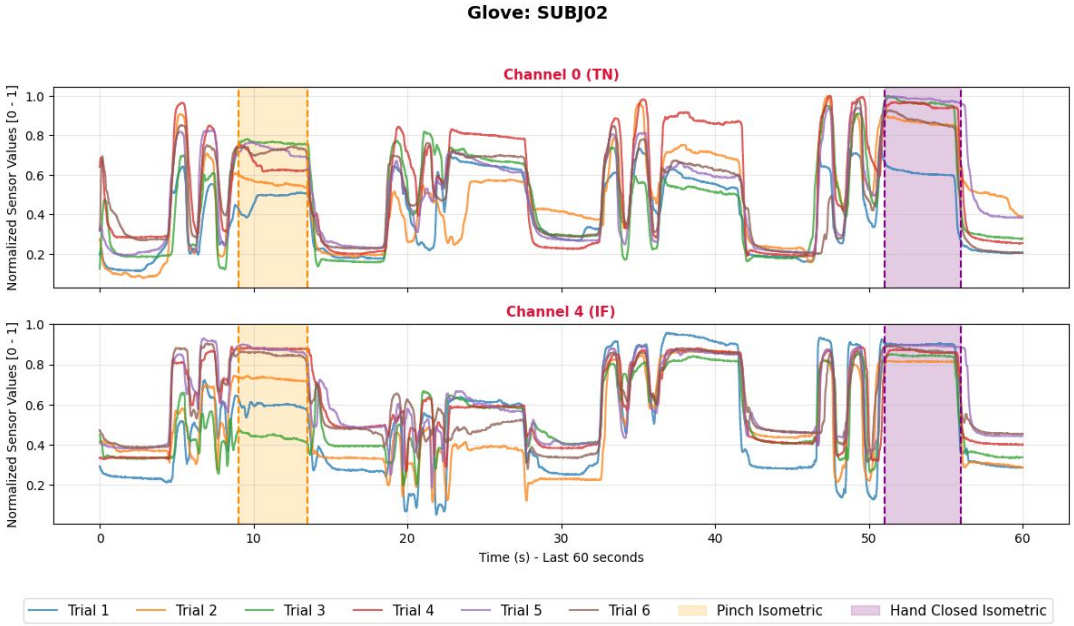
Dataset

HD FW KIN dataset

- **sEMG Signal, 3 8x8 grids**
  Fs: 2000 Hz
- **14 Channels Glove Signals**
  Fs: 200Hz
  Each represents fingers' flexion, extension, abduction.

Target Movements

- **Isometric Hand Closed**
- **Concatenated Isometric Pinch-Hand Closed**



Dataset

Far Fields Grids

- Three 8 × 8 HD sEMG electrode arrays
- 10 mm inter-electrode distance attached on the flexors/extensor muscles around forearm

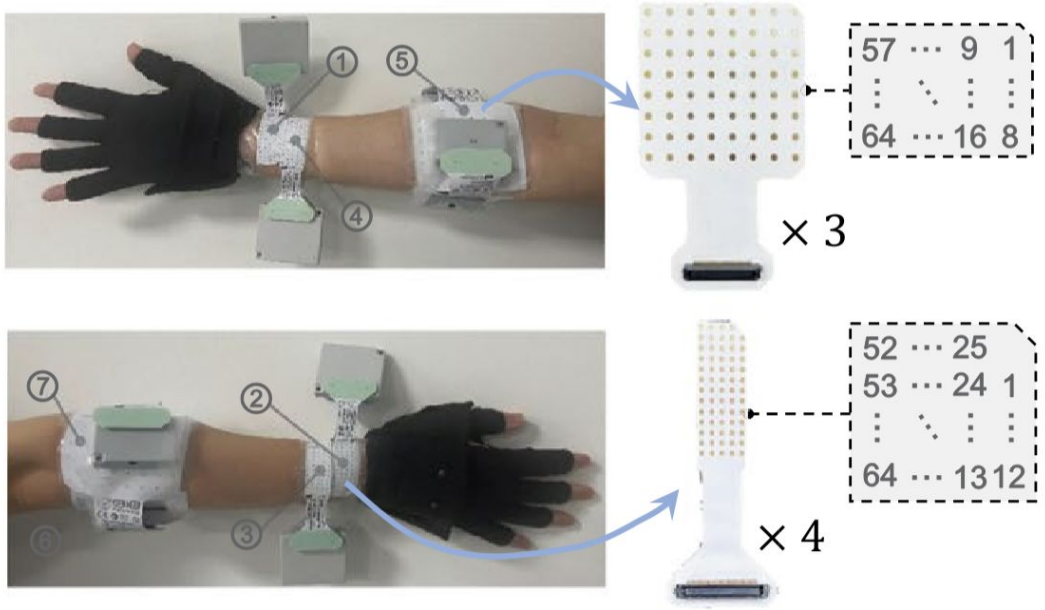
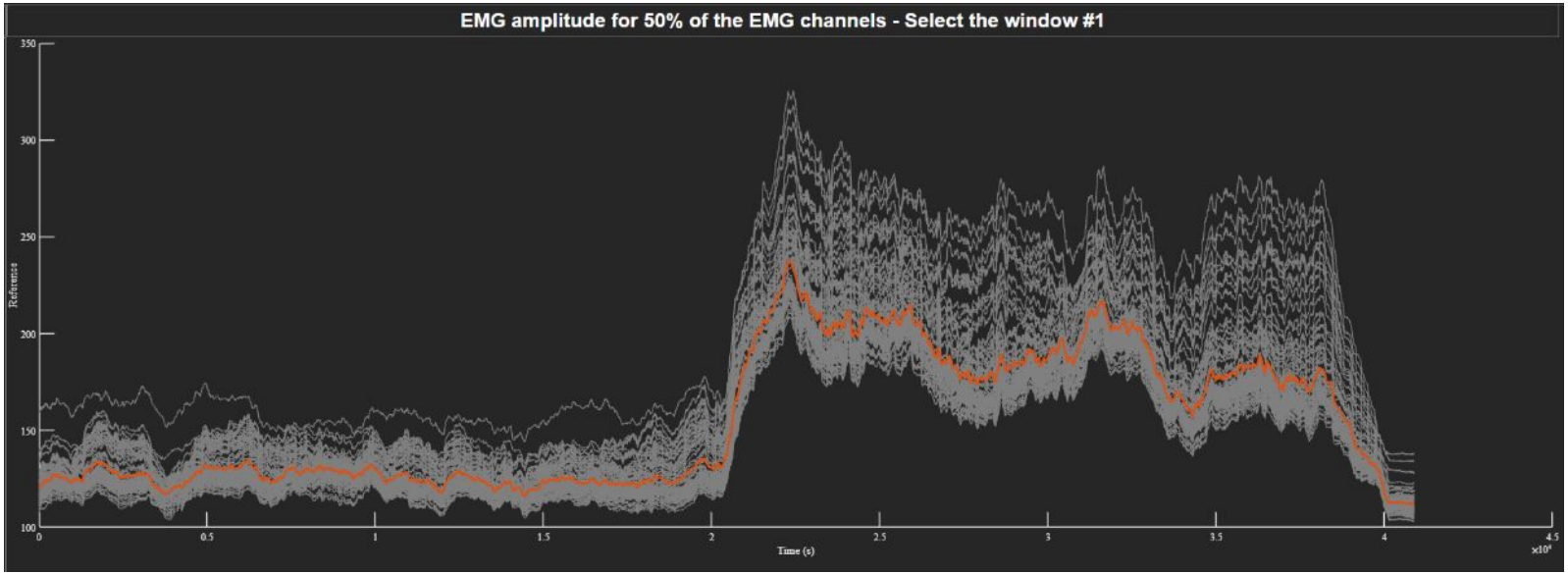


Fig. 5 “Hand kinematics, high-density semg comprising forearm and far-field potentials for motion intent recognition” Guo et al., 2025.

Preprocessing

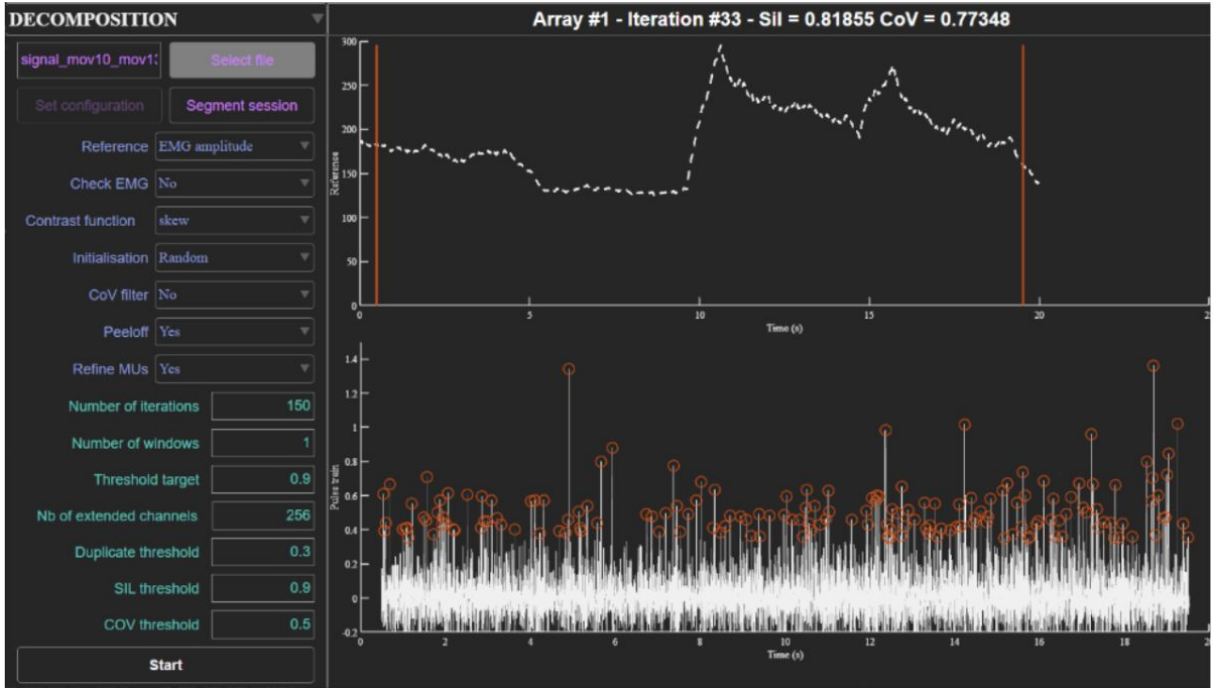
- **Band Pass Filtering**
  - 4th Order Butterworth
    20-500 Hz
- **Upsampling**
  - sEMG Signal Upsampled up to 4000 Hz and Reconstructed at Original Fs for Time Dilation



MUedit: Motor Unit Decomposition

FastICA Algorithm

- For each subject, the most isometric trial has been chosen as reference.
- MUfilters for each reference trial are saved and applied to the other trials relative to their subjects.



Postprocessing MUFilters

- **Spike Trains Discharge Rate Frequency**
  - **Each MU must have 25 spikes per task.**
- **2D normalized Cross-Correlation**
  - **For each MU, its MUAP is computed with Spike Triggered Average formulation.**
  - **MU consistency**
    Cross-correlation Threshold:
    - for MUAPs of same MU
    - different tasks.
  - **Avoid Duplicates**
    Cross-correlation Threshold:
    - for MUAPs of different MU
    - same tasks.

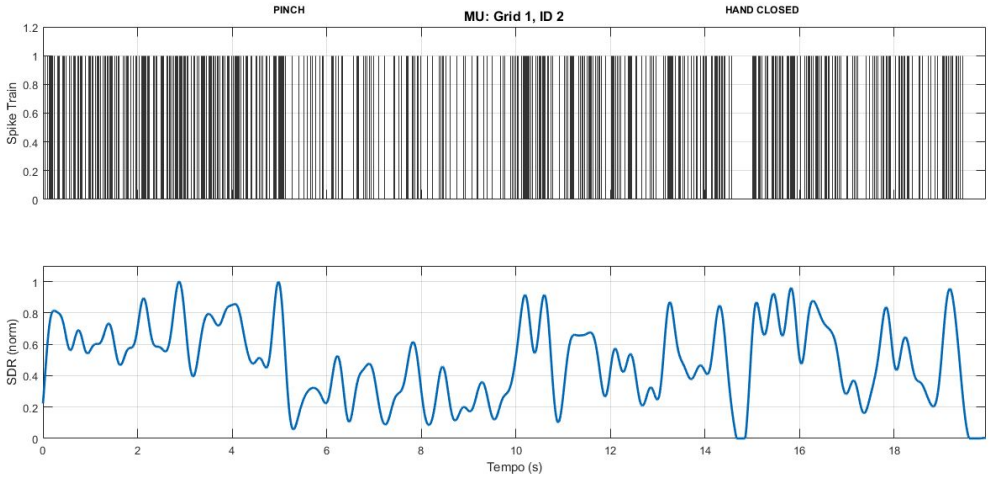


NNMF Input: SDRs Estimation

SDRs Estimation

- Low pass 4° Order Butterworth, 2.5 Hz
  - Equivalent to convolution with 400 ms Hanning window.
  - Normalization between 0 and 1

- Clipped at zero
  - to avoid negative values due to filtfilt Matlab Function

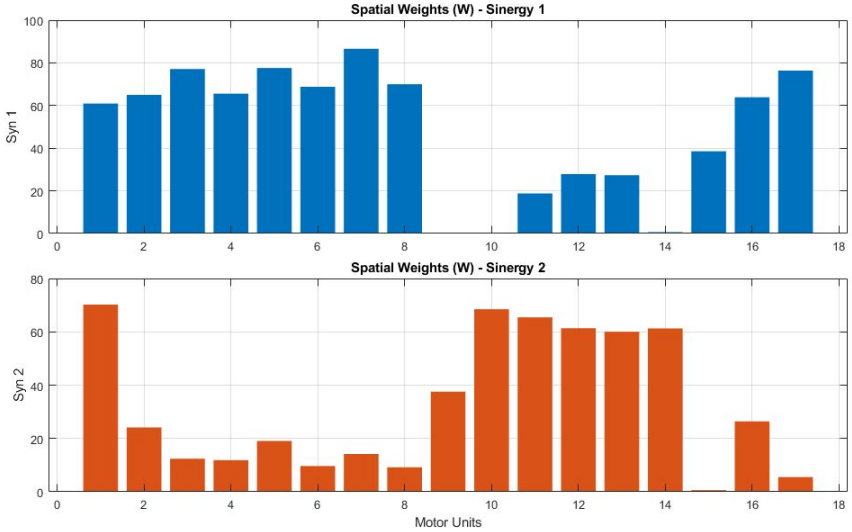


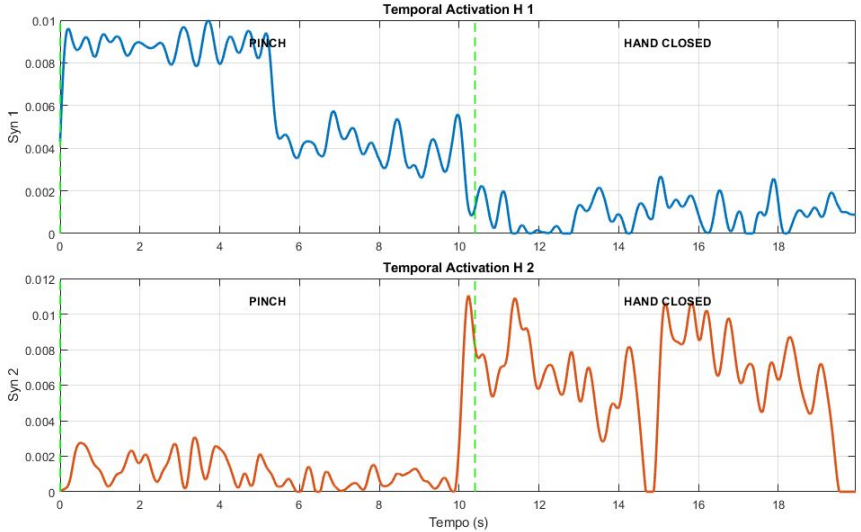
NNMF: Motoneuron Synergies

NNMF Algorithm

- Applied only on the reference trial for each subject.
- 10 Repetitions of 100 Algorithm Iterations, with *K* up to 9 synergies.
- W and H randomly initialized at each repetition.
- Number of Synergies for final Reconstruction:
  - Hand Closed only: 1 synergy
  - Pinch-Hand Closed: 2 synergies
  - Pinch-Hand Closed: 4 synergies, based on modal distribution of the Reconstruction Error (R2) across subjects.

NNMF: 2 Motoneuron Synergies Reconstruction

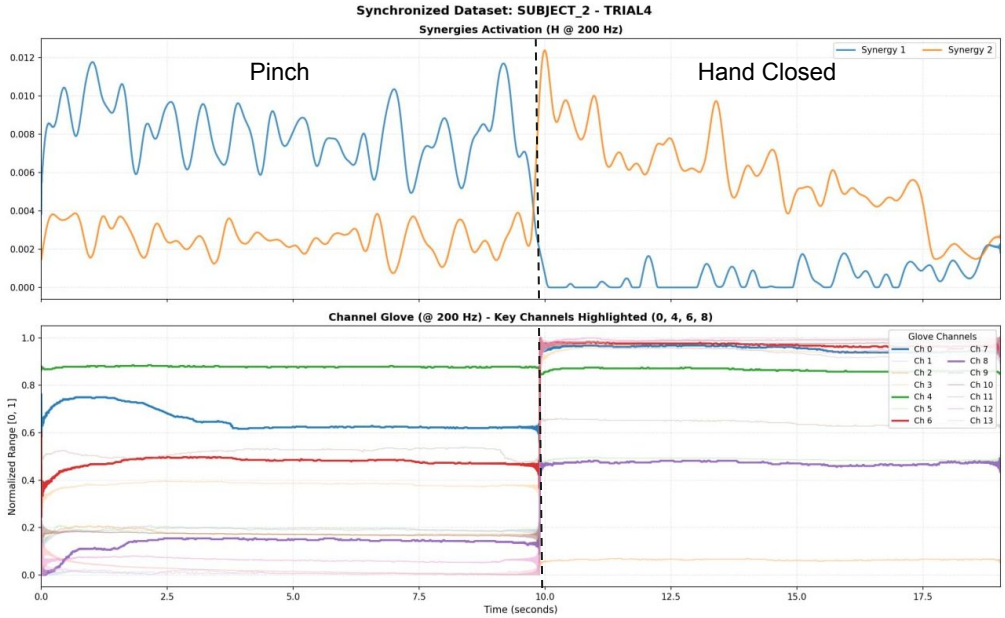




For each subject, its reference trial's Spatial Weights are used to reconstruct the other trial's SDRs.

Synchronization

- Channels Glove Upsampled up to 400 Hz and Reconstructed at its original Fs for Time Dilation.
- Downsampling of Synergies Activation, to match 200 Hz Glove sampling Frequency.
- Manual Inspection and Reordering for Consistent Synergies across subjects.

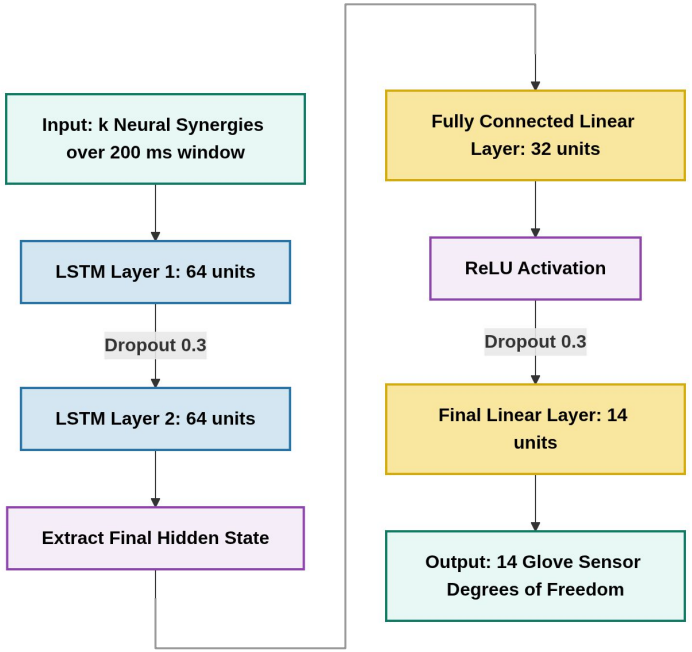


Kinematic Prediction

Training

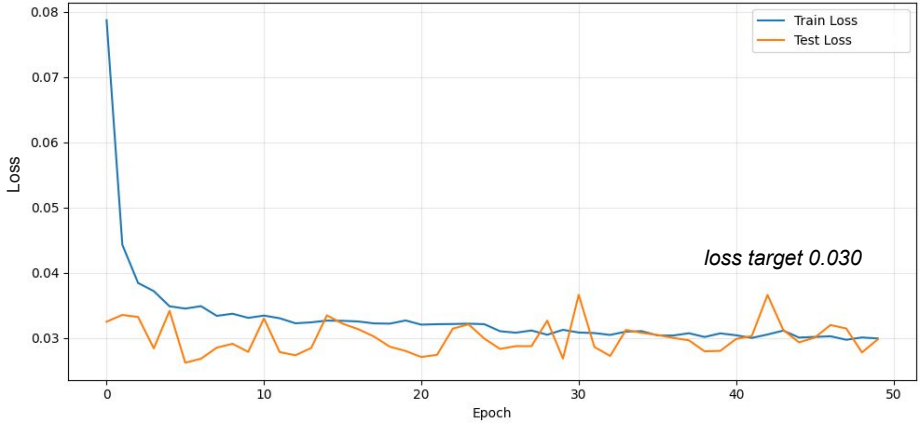
- 50 Epochs
- 200 ms Input Window (40 samples)
- Adam Optimizer:
  - LR: 1e-3
  - Weight Decay: 1e-4
- $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{smoothness}}$

- Target Prediction:
  - Intra Subject Hand Closed.
  - Intra Subject Concatenated Tasks (2 syn).
  - Cross Subject Concatenated Tasks (2 syn).
    - Training: One Subject
    - Training: Multi Subjects
  - Intra Subject Concatenated Tasks (4 syn).

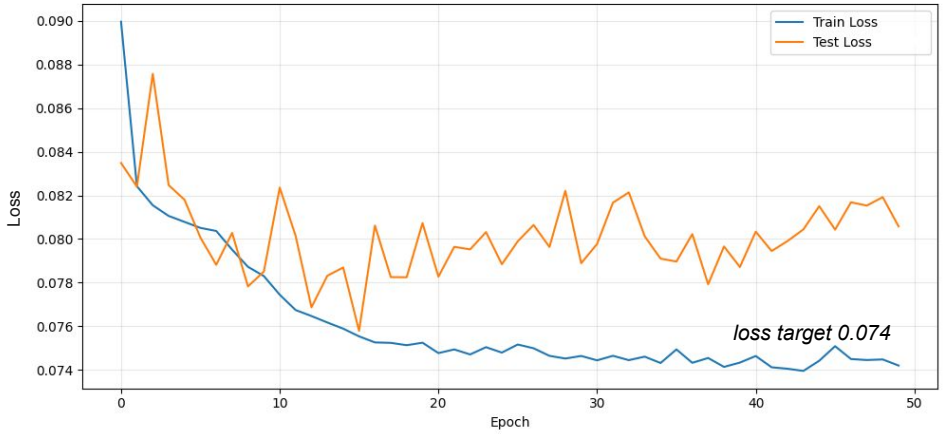


Training: 2 Motoneuron Synergies Reconstruction

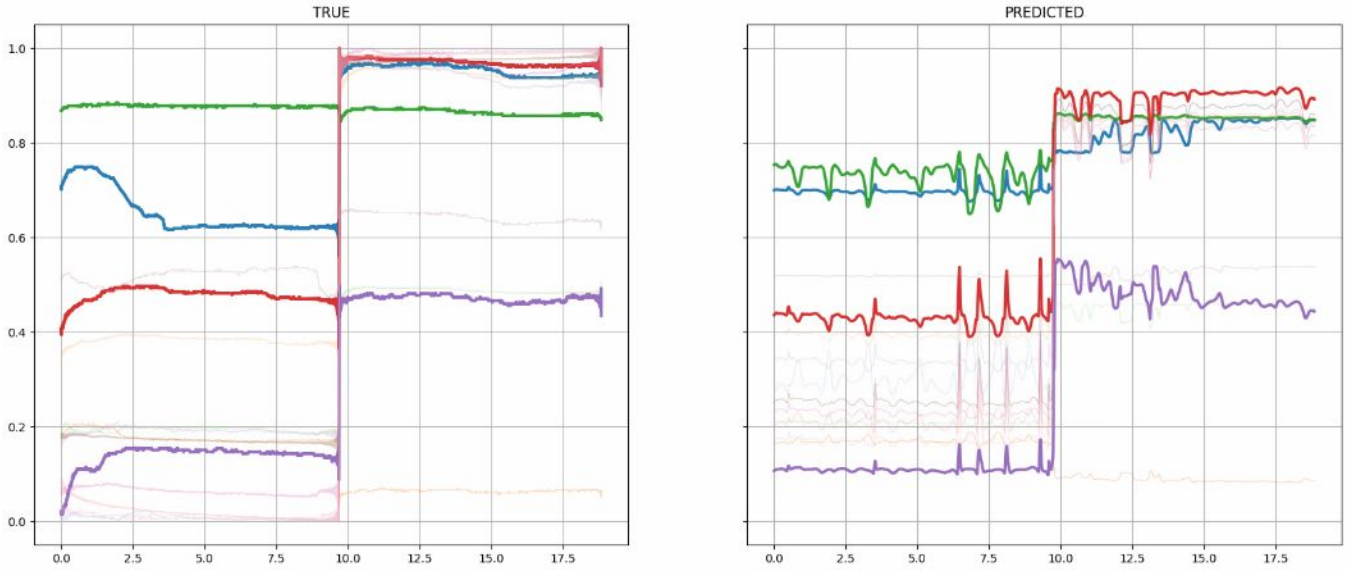
Intra Subject



Cross Subject - Multi Subject Training



Prediction: 2 Motoneuron Synergies Reconstruction



Intra Subject, prediction on unseen trial.

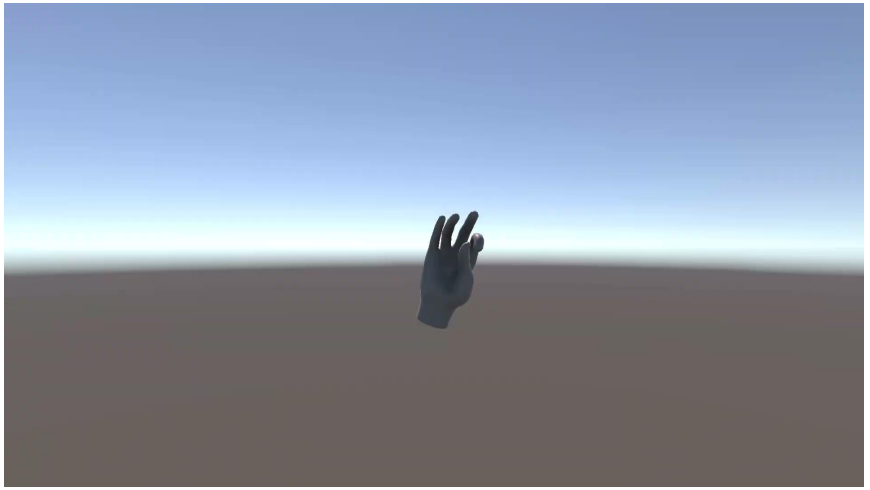
Prediction Results: Intra Subject - Hand Closed

1 Synergy

Ground Truth



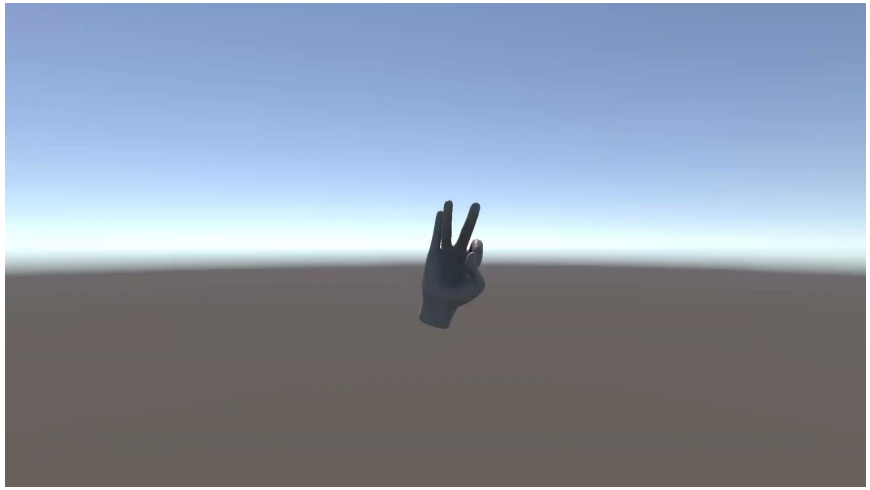
Prediction - Training with possible unstable trials



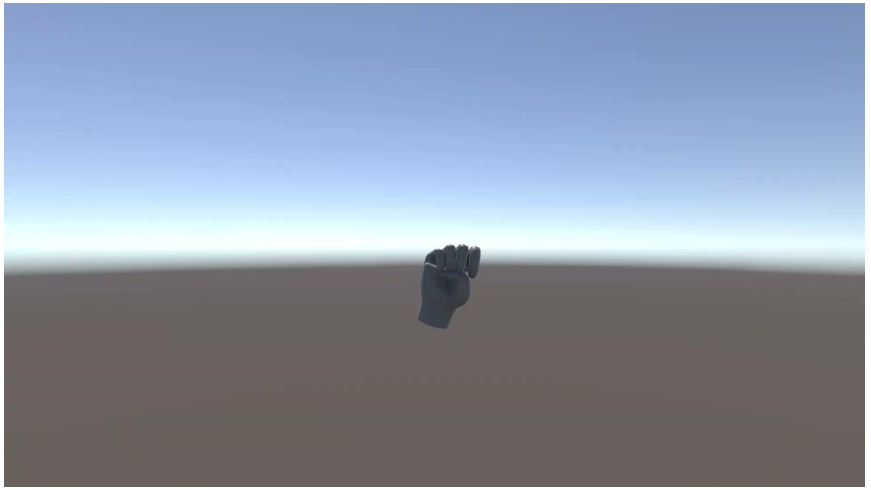
Prediction Results: Intra Subject - Hand Closed

1 Synergy

Ground Truth



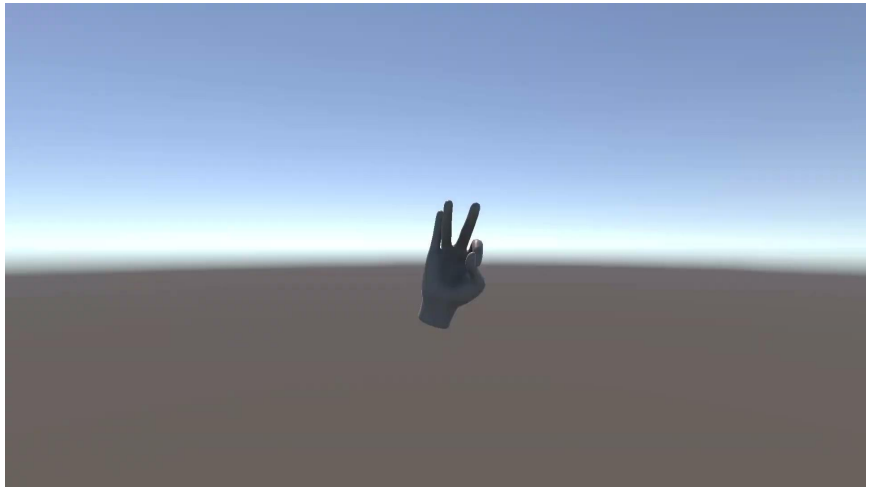
Prediction - Training on only stable trials



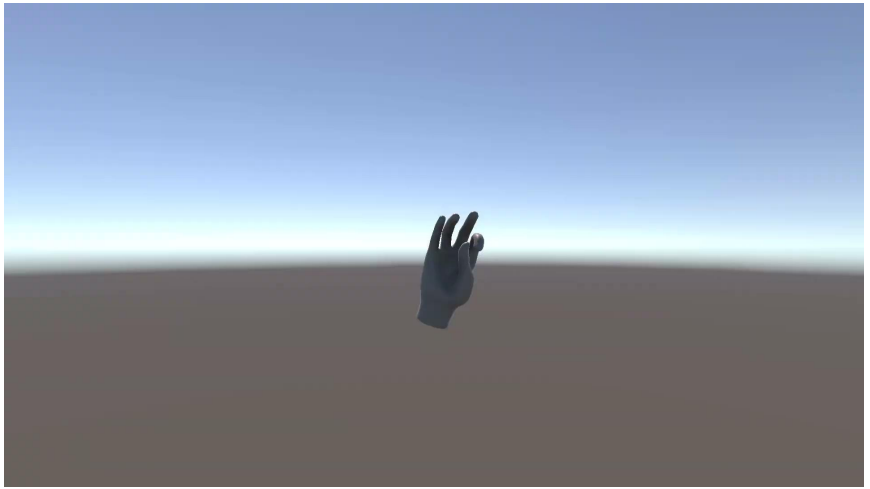
Prediction Results: Intra Subject - Concatenated Task

2 Synergies

Ground Truth

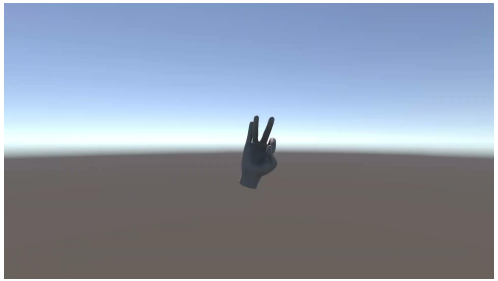


Prediction

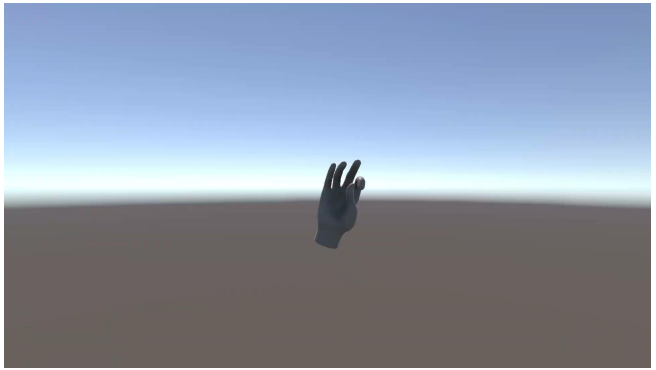


Prediction Results: Cross Subject - Concatenated Task

Ground Truth - 2 Synergies



Prediction - Training: One Subject



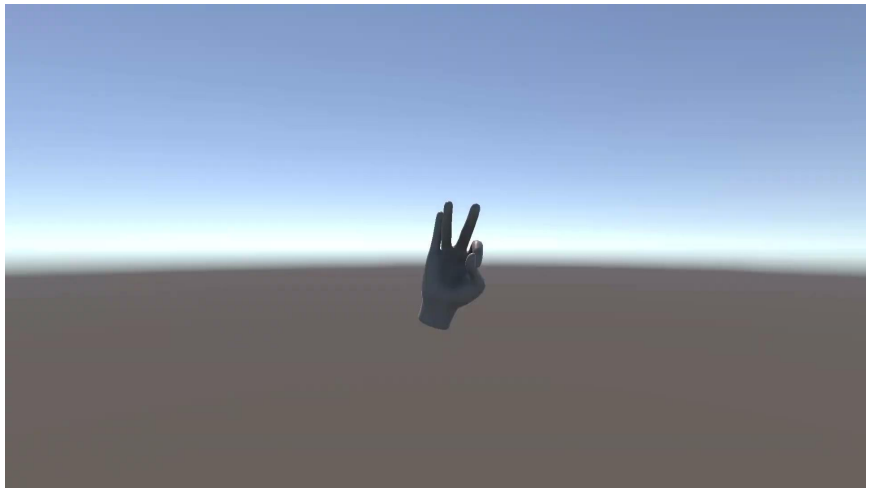
Prediction - Training: Multi Subjects



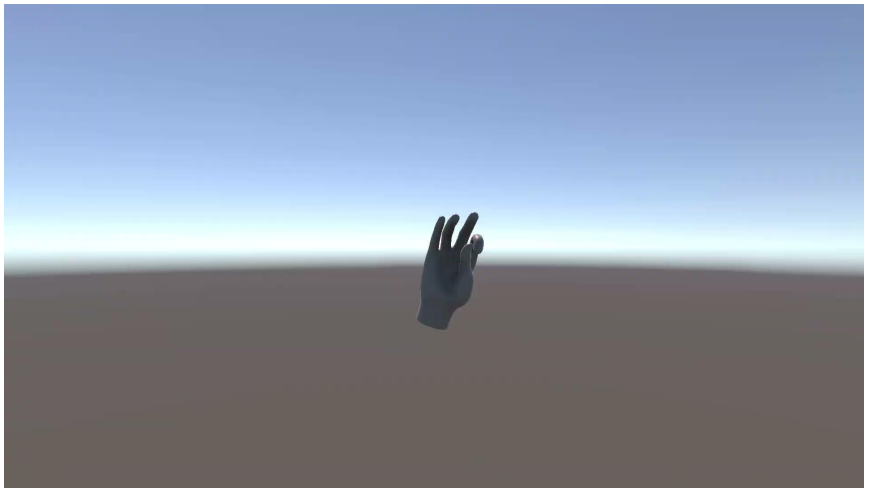
Prediction Results: Intra Subject - Concatenated Task

4 Synergies

Ground Truth



Prediction



Future Works

- Refinement of Motor Unit Decomposition and Validation
- Dataset Expansion and Synergies Correlation
- Dynamic Decomposition and Mapping
- Prediction based on Postural Synergies

Thank you for your attention!



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